FINAL HANDOFF SPECIFICATION • VERSION 1.0

Universal Financial Planning & Shock-Response Engine

Tangerine | Resilience Rewired | Sheridan Datathon 2026

Purpose: generate customer-specific financial goal and contingency plans for arbitrary shocks using Tangerine account data, transparent financial arithmetic, and measurable constraints.

CORE DESIGN DECISION Do not enumerate every possible twist. Convert any event into a small set of financial effects, then select and combine reusable financial action modules. Unknown effects stay unknown; the engine must never invent them.

1. Contract: what the product must deliver

Live flow: judge selects an account_id and describes a twist; the engine retrieves the account, displays the official current profile, applies the twist, and produces both a customer goal plan and an immediate shock-specific backup plan. There is NO requirement to simulate months of goal-plan progress before the judge's twist.

Output	Definition / judge-visible result
Baseline profile	Typical monthly money in; spending; fixed costs; household bills; cash buffer; active loan status; official runway.
Shock translation	Plain-language event -> explicit income, expense, one-off, liquidity, debt and timing effects. Show unknowns and assumptions.
Goal plan	A realistic resilience target, affordable contribution rule, and estimate of time-to-target for normal conditions.
Backup plan	What to do starting now under the selected shock; exact required gap and feasible response options.
Plan comparison	Post-shock, no-action path versus adjusted path; costs, shortfalls, horizon, conditions and trade-offs.
Explanation	Why the plan was selected; evidence and limitations from bank data; never present assumed cuts as fact.

Success criterion: for ANY customer and ANY financially interpretable twist, the system either returns a valid, coherent plan or a clearly explained unquantified/partially quantified scenario with the missing inputs. A graceful limitation is better than fabricated precision.

2. Data definitions and judge-calibrated baseline

Use the official README definitions in the judge-facing baseline. Keep separate any optional richer daily/time-varying forecast.

Symbol	Source / exact baseline definition
I	Median monthly_summary.money_in per account (money entering the account, not necessarily salary).
S	Median monthly_summary.money_out per account; includes payment categories and transfers classified as money out.
F	Median of household_bills + insurance + loan_payments for each month. Display only; NOT an extra charge in runway.
H	Median monthly_summary.household_bills.
B	Mean of avg_daily_balance over last six rows for account, month sorted ascending.
P	loans.monthly_payment if loan still running at end of 2025, else 0; reporting/loan-specific analysis.

Symbol	Source / exact baseline definition
C	One-off costs from the twist; may be zero. Do not count an expense twice.
I*, S*	Baseline I and S after applying all ongoing shock changes.

Official: buffer_left = B - one_off_costs; gap = S* - I*; runway = 0 if buffer_left <= 0; otherwise “no shortfall (120+)” if gap <= 0; otherwise min(buffer_left / gap, 120), rounded to 1 decimal. For the official EI example use I* = 0.55 × I; for bills up 15% use S* = S + 0.15 × H.

ACCOUNTING INVARIANT Do not add loan payment P to spending S if already reflected in money_out. Do not treat internal transfers between checking and savings as newly created wealth. Do not equate historical average B with verified current withdrawable cash.

3. The universal shock interpreter

Every twist - whether job loss, promotion, death, divorce, identity theft, natural disaster, investment crash, inheritance or compound shock - must be decomposed into the same financial primitives. One event can have several primitives.

Primitive	Parameter examples	How it changes the financial projection
Ongoing inflow	-45% inflow for 6 months; +$600/mo from month 2	Change I_t by effective date and duration.
Ongoing outflow	+$250/mo rent; +15% household bills	Change S_t, avoiding duplication of existing expense categories.
One-time cash event	-$3,200 repair on day 0; +$8,000 gift	Change available cash only when amount is received/paid.
Liquidity restriction	-$2,000 inaccessible for 30 days	Reduce spendable liquidity; distinguish temporary hold from permanent loss.
Debt/service change	New $180 monthly payment; authorized deferral	Change actual payment schedule and explicitly model fees/costs.
Investment/asset value	Portfolio -25%; stock sale net of fees/tax	Change ONLY verified investment asset, not bank cash by default.
Timing change	Deposit delayed 20 days; bill moved to 5th	Change dated cash-flow sequence, not necessarily monthly totals.
Conditional coverage	Existing insurance pays $2,500 if eligible	Only include when policy, trigger and amount are confirmed or openly assumed.

Proposed event object (financial contract, not an ML choice)

Fields: event_id; narrative; primitive_type; affected_account_component; signed_amount or percent; percentage_base; start_date/relative_start; duration_months or one_time_date; recurrence; eligibility_status; confirmed_vs_assumed; confidence; financial_effects; question_needed. Multiple effects per twist are allowed.

Example: “laid off and bills rose 15%” -> [income multiplier 0.55 for months 1-6, household_bill_cost +0.15 × H starting month 1]. A third event such as a $1,500 repair is simply appended to the same scenario.

Robust interpretation rules

- Apply absolute changes and percentage changes against a clearly identified original base; document order if multiple percentages target the same field.
- Keep positive shocks (raise, inheritance, forgiven debt) as first-class cases, not errors; then generate an allocation/opportunity plan.
- If shock duration is unknown, display an immediate/month-1 case and a configurable horizon, with duration labelled as an assumption.
- A vague event without a financial effect (“someone did a backflip”) produces no change; if injury results, ask for costs and income effects or offer editable hypothetical values.

- A death, divorce or illness is NEVER assigned assumed dollar amounts based on demographic stereotypes; request income loss and actual obligations.
- A natural-language model may extract fields, but a deterministic financial calculator performs the arithmetic and checks invariants.
- If unknown inputs remain, calculate the known portion and show a range or “not estimable” instead of making up salary, coverage or expenses.

4. Financial model: month-by-month projection

The one-number official runway is a calibration check. For plans lasting multiple months, use a time-indexed cash model to handle expiring EI, delayed income, one-time costs, seasonal costs and combinations.

For month t: projected_cash[t] = projected_cash[t-1] + inflow[t] - outflow[t] + verified_cash_receipts[t] - verified_one_off_costs[t] + net_realized_financing[t]. Begin with a separately labelled proxy B (or verified accessible opening cash if supplied). Do not count the same transaction twice.

Liquidity available[t] = projected_cash[t] - funds_held_or_inaccessible[t] - amounts reserved for protected near-term obligations. A projected cash shortfall occurs when this quantity falls below zero. If a financing action adds cash, its repayments, interest and fees must appear in later periods.

Useful outputs: first shortfall month or day; minimum projected balance; total shortfall over target horizon; six-month and twelve-month remaining cash; official runway; action-adjusted runway; unresolved funding gap. Monthly projection is sufficient for the minimum version, with daily timing as optional enhancement.

Plan optimization objective (lexicographic, not a magic score)

1. 1. Preserve minimum essential obligations and avoid impossible or illegal actions.
2. 2. Eliminate or minimize immediate funding gaps and projected negative liquidity.
3. 3. Maximize survival through the shock’s actual duration (default demonstration: six months when duration unspecified).
4. 4. Minimize total added financial cost, forced asset sale, hardship, and irreversible commitments.
5. 5. Once stable, rebuild reserve and improve longer-term financial position.

A plan may fail to fully close a gap. Return the best feasible partial response and the exact remaining deficit. Never fabricate available discretionary spending.

5. Reusable response plan library (8 modules)

These are mechanisms, NOT a lookup table of specific twists. Select a lead module based on the dominant funding problem; combine supporting modules when needed.

Code / Module	Trigger pattern	Calculated response
M1 Income Continuity	Recurring income reduction, delayed wages, gig-work interruption	Monthly deficit, support-eligibility scenarios, affordable reduction target, reserve draw schedule.
M2 Cost Rebalance	Permanent/temporary expense rise, rent, childcare, inflation	Incremental essential-cost burden; adjust only verified flexible items; new sustainable surplus.
M3 Emergency Liquidity	One-off repairs, emergency expense, theft	Immediate cash gap, accessible reserves, verified coverage, approved staged payment, recovery timeline.
M4 Debt & Obligation Defense	Debt service threatened, interest or payment shock	Payment-to-cash-flow stress; preserve essential obligations; price approved hardship/refinance options.
M5 Multi-Shock Stabilization	Several shocks together or survival deficit severe	Combine modules; rank necessary actions; identify unresolved amounts and shortfall date.
M6 Reserve Restoration	Cash buffer depleted, shock ends, monthly surplus returns	Accessible target, affordable contribution rate, time to restore; optional liquidity tiers.

Code / Module	Trigger pattern	Calculated response
M7 Timing & Access	Delayed deposits, uneven cash flow, frozen funds	Daily/weekly payment calendar; timing adjustments where allowed; temporary liquidity bridge.
M8 Surplus & Windfall Allocation	Raise, inheritance, bonus, debt forgiveness	Priority order: essentials/arrears -> reserve -> high-cost debt -> optional longer-term goals.

Universal action primitives used inside modules

A01 pause optional outflows; A02 reduce customer-confirmed adjustable spending; A03 use verified available cash; A04 activate existing eligible insurance; A05 check income support/benefits; A06 seek paid income/temporary support (not guaranteed); A07 request bill-date change; A08 seek lender-approved hardship adjustment; A09 compare payment plan/credit only if eligible and total costs shown; A10 restore reserve; A11 allocate excess to protected cash savings; A12 consider investments only after core liquidity needs; A13 release verified held funds; A14 sell a verified liquid asset net of losses/tax/fees; A15 update protection assumptions; A16 present unresolved gap and external-assistance route.

Plan selection rule

Classify the actual problem, not the person's income bracket alone. First detect liquidity constraints (immediate gap? recurring gap? one-off gap? timing issue? positive surplus?). Then select the lead module and include up to two supporting modules. Never use income or gender as a substitute for proof of feasibility.

6. Outputs: goal plan vs. shock backup plan

Goal plan (generated at runtime)	Backup plan (generated at runtime)
Normal-case target, e.g. essential-expense reserve / scenario-specific reserve.	Selected event(s), magnitudes, duration, and modified financial profile.
Required additional reserve = max(0, target - verified accessible reserve).	Post-shock immediate gap, monthly gap and projected first shortage.
Contribution schedule from demonstrated stable free cash, with feasibility check.	Ordered actions with amounts, eligibility, costs, implementation assumptions.
Target time and reason the contribution rule fits income variability.	New projected runway and balance path only for actions marked confirmed/assumed appropriately.
If unaffordable, no artificial timeline; offer smaller milestones or stabilization first.	Report remaining uncovered gap; recovery actions when the shock resolves.

These two plans are produced together AFTER entering account_id and the judge's twist. The goal plan is a baseline resilience objective; the backup plan addresses the hypothetical shock immediately. Neither requires a historical simulation of the customer having already completed the goal plan.

Required human-readable recommendation fields

- One-line diagnosis ("Your monthly inflow falls by $1,800").
- Official runway and formula basis ("4.0 months based on $4,000 / $1,000").
- Specific action and quantity ("An additional $333.34/month is needed to cover six months").
- Feasibility ("We cannot verify $333.34 of adjustable spending from cash-withdrawal data").
- Result ("If $350/month is actually reducible, simulated runway becomes 6.2 months").
- Alternatives and trade-offs ("Seek coverage/support if eligible; new credit adds cost").
- Missing facts, confidence labels and unresolved risk ("Income duration assumed six months").

7. Optional insurance, investing and product innovation

These elements can differentiate the submission, but they are SCENARIO OVERLAYS, not free money or guaranteed customer benefits.

Product idea	What is legitimate in the simulator	Hard restriction
Existing insurance benefit	Add a payout only if covered event, policy and payout schedule are known or explicitly hypothetical.	Do not assume buying insurance after a loss makes that past loss insurable.
Protected reserve tiers	Separate immediately accessible cash from higher-yield savings or investment assets.	Only verified accessible funds count toward immediate survival.
Investment return/ROI	Model returns as scenarios, including severe losses and liquidity delay.	Do not count stock market appreciation as certain emergency cash.
Credit/payment bridge	Model amount available, eligibility, fees, interest and later repayments.	Do not recommend unapproved debt as a guaranteed solution.
Risk pooling / mutual reserve	Compare hypothetical pooled contributions and losses in a research-only scenario.	Do not claim insurance pricing or regulatory eligibility is validated by bank data.

Creative product position: “Financial Mission Control” - the engine tests whether cash, expense adjustments, existing protection and other feasible actions can absorb shocks, then explains the cheapest robust response. The innovation is in the comparison and defensibility, not the naming.

8. Worked demonstration scenarios

Hypothetical customer: I = $4,000/month; S = $3,200/month; H = $1,200/month; B = $4,000. Normal surplus is $800/month. Values are demonstration assumptions, not customer data.

Twist	Official effect / initial result	Primary response
Job loss; EI 55%	I*=2,200; S*=3,200; gap=1,000; runway=4.0 months.	M1. Six-month gap is $2,000 beyond buffer; need $333.34/month effective improvement. Confirm cut feasibility.
Repair costs one month of spending	C=3,200; remaining buffer=800; ongoing gap <= 0; official “no shortfall (120+)”.	M3 + M6. Show cash depletion; $400/month verified reserve contribution would restore $3,200 in 8 months.
Job loss + 15% household bills	I*=2,200; S*=3,380; gap=1,180; runway=3.4 months.	M5 + M1 + M2. Need $513.34/month effective improvement to cover six months.
Salary rises 10%	I*=4,400; S*=3,200; surplus=1,200/month.	M8. Allocate incremental $400/month according to reserve/debt priorities, not automatic risky investing.
One-time $10,000 inheritance	Cash +10,000 only when received; no ongoing monthly shock.	M8. Resolve arrears, protect liquidity, compare debt reduction and long-term saving; no fabricated guaranteed return.
Cash account holds $2,500 for 2 months	Historical B remains 4,000 in judge baseline; accessible proxy temporarily 1,500.	M7. Evaluate near-term due dates, verified alternate liquidity and hold expiry; don’t treat hold as permanent theft.
Emergency $6,000 bill today	C=6,000; buffer_left=-2,000 => official runway=0.	M3 + M5. Immediate uncovered gap of $2,000; no pretend reimbursement or guaranteed loan.
“Someone backflips” without other details	No established financial effect -> no change to I, S or B.	Return unchanged baseline and request injury/cost details if relevant. Do not guess.

9. Edge cases and fail-safe policy

Case	Required behavior
Unknown account ID / no rows	Display “Account unavailable”; no derived quantities; skip action optimization.
Fewer than six monthly rows	Use available rows per README; visibly show history length and uncertainty.
Income zero / negative typical surplus	Do not generate unaffordable mandatory saving targets. Prioritize stabilization.
Negative or zero buffer	Official runway = 0 when buffer_left <= 0; do not show negative months.
S* <= I*	Official “no shortfall (120+)”, but still evaluate large one-off depletion and short-term timing risk.
Shock amount negative / percentage >100%	Validate signs and scope; reject physically impossible salary values unless explicitly representing debt/negative income.

Case	Required behavior
Multiple and conflicting effects	List all effects and calculate from known bases; avoid double-counting related rent/bill increases.
Temporary EI or benefit expires	Model changes when benefit duration ends, not as permanent income.
One-off cost exceeds cash	Immediate uncovered deficit; any additional financing conditional on confirmed availability.
Loan already inside spending	Never append P to S unless an explicitly new payment is added.
Investments not in data	Default value = unknown/zero for available cash; only user-entered investment positions may be simulated.
Sensitive life events	Do not infer diagnosis, insurance eligibility, household earners, or expenses from age/gender.
Unclassifiable twist	Return financial primitives found, missing parameters and an editable assumption panel. No hallucinated plan.

10. Data provenance and limitations

- Tangerine data: 4,500 accounts, monthly_summary (185,326 rows), transactions (1,056,320 rows), loans (682), and related relational tables. Recalculate real counts from files before final pitch.
- Income proxy: money_in aggregates cash deposits, transfers and other receipts; it is not an employment salary field. “55% of usual income” is the judge’s specified scenario convention, not proof of EI eligibility or actual benefit amount.
- Expenses: withdrawals are not reliably classifiable into groceries, recreation, etc. A separate preliminary research note claims ~76% of outflow is cash withdrawals; reproduce before citing.
- Buffer B is a historical measure, not current savings balance or verified cash accessible today. Label short-term liquidity stress tests as optional extensions rather than the official baseline.
- The data contains loan payments/status but not complete external liabilities, insurance policies, income-support eligibility, current portfolio values, tax position or risk tolerance.
- Claims about financial outcomes are simulated counterfactuals. Do not say a customer WILL make suggested cuts, receive benefits or be approved for credit.

11. Testing and competition-quality evaluation

Core acceptance test matrix

Test family	Minimum tests
Official arithmetic	Median definitions, B last-six average, EI 55%, 15% bills, one-off costs, zero gap, cap 120, rounding.
Twist generalization	Raise; wage loss; inheritance; death with user-given effects; divorce; accident; loan spike; inflation; account hold; delayed paycheck; natural disaster; null shock.
Combinations	Job loss + rent increase + one-off repair; multiple dates; mixed percent and absolute; overlapping times.
Financial consistency	No duplicated loan costs; no transfer-based wealth creation; no insurance payout without coverage; debt payments/fees after credit.
Feasibility	Savings plan unaffordable; no reducible spending; zero/negative cash; inaccessible buffer; unfunded cost.
Interpretability	Every action linked to input/assumption; missing information shown; no unsupported customer claims.
Performance	Random account + random event renders before live explanation; computations cached only if all relevant scenario inputs match.

Evaluation methodology

- Check official benchmark calculations for a sample of accounts and every organizer example. Require exact equality after matching their rounding.

- Run event/property-based tests on arbitrary account numbers and shock magnitudes; check that larger costs do not spuriously increase runway (all else equal).
- Historical test of savings affordability: use earlier months to form a contribution rule and later months to measure reversals, negative balances and progress.
- For shock response, compare no action versus suggested actions under identical scenarios; measure funding-gap reduction, cost and feasibility. Do not claim causal customer outcomes.
- Perform a demonstration stress test: random account, odd natural-language event, multiple shocks, missing values, and a positive windfall.

Competitive discovery to investigate

Hypothesis: typical monthly inflow/spending and average balances hide cash-flow timing and consistency problems. Investigate last-12-month deficit frequency and negative-balance days; compare similarly sized accounts so the finding is not simply “wealthier customers have more money.” The separate research note reports 714/3,309, 1,986/2,902, and 612/1,326 illustrative findings. These require independent reproduction before being used as evidence.

12. Build ownership and priority

Priority	Finance team deliverable	Engineer integration contract
P0: must ship	Canonical formulas; 8 module definitions; shock primitive library; feasibility rules; 8 worked examples.	Deterministic calculator taking structured event list and customer features.
P0: must ship	Judge-facing output fields and explanation wording.	Render primary plan, backup plan, no-action vs action comparison and caveats.
P1: strong differentiator	Month-by-month cash paths, intervention optimization, one confirmed alternative to baseline.	Planner searches feasible action combinations and returns proof of selection.
P1: strong differentiator	Reproduce 1-2 discovery findings and honest limitations.	Discovery page with denominators and filters.
P2: if time	Daily liquidity timing, insurance/investment sandbox, ML clustering or NLP shock parsing.	Optional add-ons; cannot break P0 live demo.

Minimal plan output data contract

account_id; history_months; baseline {I,S,F,H,B,P,runway}; shock {narrative,effects,assumptions,unresolved_inputs}; forecast {month,base_cash,action_cash,accessible_cash,gap}; goal_plan {target,contribution_rule,target_date,feasible}; backup_plan {lead_module,support_modules,actions[],verified_vs_hypothetical,post_action_runway,remaining_gap,tradeoffs}; explanation {drivers,limitations,confidence}.

FINAL DECISION Universal coverage is achieved through composable financial effects and actions, NOT through an impossibly long list of pre-written twists. Guarantee that any event can be represented, partially quantified or transparently flagged - not that the engine knows facts absent from the banking data.

Source notes

Primary specification: supplied “Tangerine / Resilience Rewired” README (Sheridan Datathon 2026), including official metric definitions and judge twist examples. Supporting input: user-provided financial research note with preliminary, unverified descriptive findings. External product rules, insurance contracts, interest rates and eligibility are not provided; all such features are hypothetical until verified.

This is a project design specification, not individualized regulated financial or insurance advice. Any client-facing product would need eligibility, consent, governance, compliance and additional financial inputs.